

- 5 (a) When two or more waves of the same kind meet at a point at the same time, **B1**
the displacement of the resultant wave is the vector sum of the displacements
of the individual waves at that point at that time. **B1**

(b) (i) $\lambda = \frac{v}{f} = \frac{330}{200} = 1.65 \text{ m}$ **A1**

- (ii) The distance of the upper loudspeaker from the man is

$$\sqrt{9.3^2 + 4.0^2} = 10.124 \text{ m}$$

$\therefore$ the path difference between the waves reaching the man is

$$10.124 - 9.3 = 0.8237 \text{ m}$$

C1

the path difference is $\frac{0.8237}{1.65} = 0.50\lambda$

C1

This means that the waves interfere destructively and the man will hence detect
a **minimum** of sound intensity. **A1**

- (iii) 1. increase the frequency of the sound

B1

2. increase the separation between the two loudspeakers

B1

6 (a) The electric force per unit positive charge experienced by a small stationary